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Apolipoprotein AI_{Milano}

DISULFIDE-LINKED DIMERS INCREASE HIGH DENSITY LIPOPROTEIN STABILITY AND HINDER PARTICLE INTERCONVERSION IN CARRIER PLASMA*

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The *in vitro* metabolism of high density lipoproteins (HDL) in carriers of the apolipoprotein AI_{Milano} ($apoAI_M$) mutant was investigated during incubation of whole plasma and isolated lipoprotein fractions. A reduced cholesterol esterification (16.5 *versus* 25.0% for controls) and a decreased exchange of lipids between HDL and lower density lipoproteins was observed during incubation (6 h at 37 °C) of AI_M plasma. Control HDL_3 were converted to larger, faster-floating HDL particles, whereas only a fraction of AI_M HDL_3 followed the same pathway. Incubations were also carried out by mixing HDL_3 from controls and AI_M carriers with a lipoprotein-depleted plasma fraction in the presence of triglyceride-rich particles isolated from Intralipid. AI_M HDL_3 again showed a reduced capacity for lipid exchange; some HDL_3 particles followed a "normal" conversion to faster-floating, larger HDL, whereas the small AI_M HDL_3 were not modified, indicating that AI_M HDL_3 are a mixture of metabolically functional and nonfunctional particles. Following transformation of the $apoAI_M$ homo- and heterodimers into their normal counterparts, *i.e.* monomeric apoAI and -AII, by reduction and carboxamidomethylation of AI_M HDL_3, the modified HDL_3 behave like control HDL_3 during incubation with lipoprotein-depleted plasma and triglyceride-rich particles. The presence of AI_M dimers is most likely responsible for the increased HDL_3 stability in the AI_M carriers, indicating that apolipoprotein composition plays a major role in HDL particle interconversion.

Human high density lipoproteins (HDL)[1] represent a family of particles differing in size, density, lipid, and apolipoprotein content (1). Based on density gradient ultracentrifugation, two main subfractions can be separated, identified as HDL_2 (d = 1.063–1.125 g/ml) and HDL_3 (d = 1.125–1.21 g/ml) (2); each of these major subfractions can be further fractionated into distinct entities by different techniques (3, 4).

The metabolism of HDL is predominantly intravascular, HDL particles being continuously converted in plasma (5, 6). This process consists of several steps, involving different plasma activities, *i.e.* lecithin:cholesterol acyltransferase (LCAT), various lipid transfer proteins (LTP), lipases, and conversion proteins (7, 8). The end result is a reverse transport of cholesterol from the periphery to the liver (9). It is generally accepted that smallest pre-β migrating HDL containing only apoAI are the most active particles in promoting the transfer of cholesterol from cell membranes to the plasma compartment (10, 11). Once incorporated into these small particles, the free cholesterol is esterified by LCAT (9); low density lipoproteins are also known to supply free cholesterol and phospholipids to HDL for the LCAT-catalyzed reaction (12), in a process mediated by a specific phospholipid transfer protein (or LTP-II) (13). The further addition of surface material derived from the catabolism of triglyceride-rich lipoproteins converts HDL_3 to HDL_2-like particles (14). A second lipid transfer protein (LTP-I or CETP) transfers the newly formed cholesteryl esters (CE) to very low density lipoproteins in exchange for triglyceride (TG) (15). This results in the formation of CE-poor, TG-enriched HDL_2, that can be converted back to small HDL_3 by the action of hepatic lipase (16).

Clinical syndromes characterized by a deficiency of one of the several factors involved in HDL particle interconversion are associated with significant abnormalities in the HDL_2-HDL_3 distribution. LCAT deficiency results in the accumulation of small HDL_3 in plasma (17); hepatic lipase deficiency is associated with a relative abundance of TG-rich HDL_{2b} (18) and CETP deficiency leads to a hyperalphalipoproteinemia, characterized by the accumulation of CE-rich HDL_2 particles (19).

The carriers of the apolipoprotein AI_{Milano} ($apoAI_M$) mutant (20, 21) show multiple abnormalities in the structure and composition of HDL. HDL_2 are nearly absent and HDL_3 are heterogeneous, compared to normal subjects (22, 23); all these particles are generally TG-enriched and CE-poor. The heterogeneity in the HDL pattern has been related both to the presence of the $apoAI_M$ homo- and heterodimers (22) and to the plasma HDL-cholesterol (HDL-C) and TG levels (23). The LCAT deficiency state recently detected in the AI_M carriers (24) can also play a role in the formation of the anomalous HDL found in these subjects. Since the HDL

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[1] The abbreviations used are: HDL, high density lipoproteins; $apoAI_M$, apolipoprotein AI_{Milano}; CE, cholesteryl ester; CETP, cholesteryl ester transfer protein; GGE, polyacrylamide gradient gel electrophoresis; HDL-C, HDL-cholesterol; LCAT, lecithin:cholesterol acyltransferase; LPDP, lipoprotein depleted plasma fraction; LTP, lipid transfer proteins; PL, phospholipids; RCM, reduced-carboxamidomethylated; TC, total cholesterol; TG, triglyceride; TGRP, triglyceride-rich particles purified from Intralipid.

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pattern is a result of the complex interplay of LCAT and LTP activities with donor and acceptor lipoproteins, we evaluated the *in vitro* effects of endogenous and exogenous LCAT and LTP on the distribution, composition, and structure of HDL subfractions in the AI_M carriers.

EXPERIMENTAL PROCEDURES

Subjects—Studies were performed on plasma samples obtained from different AI_M carriers and from age-sex-matched controls from the same large kindred (25). Hypertriglyceridemic control subjects were selected among those attending the Center E. Grossi Paoletti Lipid Clinic; they were on a standard, low-lipid diet and free from hypolipidemic drugs for at least 3 months. All participating subjects were fully informed of the modalities and end points of the study, which was approved by the Internal Review Board. Blood from subjects fasted 14 h was collected in Na_2EDTA (1 mg/ml) and placed immediately on crushed ice. Plasma was prepared by low speed centrifugation at 4 °C, and incubation and/or lipoprotein isolation were started within 6 h after blood collection.

Isolation of Lipoprotein Fractions—HDL (d = 1.063–1.21 g/ml) and HDL subclasses were separated either by salt density ultracentrifugation (26), or by rate zonal ultracentrifugation (27), or by nondenaturing polyacrylamide gradient gel electrophoresis (GGE) (3). Salt density ultracentrifugation was carried out in a Ti-50.2 rotor; lipoprotein fractions were isolated and washed at their upper densities for 40 h at 45,000 rpm at 4 °C and stored under N_2 at 4 °C. Zonal ultracentrifugation was performed at 15 °C using a Beckman SW Ti-41 rotor (27): samples were separated by a nonequilibrium rate zonal ultracentrifugation at 40,000 rpm for 23 h and the tube effluent monitored continuously at 280 nm. The elution volumes (V_e) of the major peaks were automatically recorded by a Pharmacia LKB Biotechnology Inc. FRAC-100 fraction collector (27). Ultracentrifugal d < 1.20 g/ml fractions were isolated in a Beckman TL-100 ultracentrifuge equipped with a Beckman TL-100.3 rotor at 100,000 rpm for 3.5 h at 4 °C and analyzed by GGE to determine the particle size distribution of HDL. After staining, gels were scanned by a laser densitometer (Ultroscan, LKB, Bromma, Sweden) and HDL particle size determined as described previously (23).

A plasma fraction depleted of lipoproteins (LPDP) by ultracentrifugation at d = 1.215 g/ml (45 h at 45,000 rpm in a Beckman Ti-50.2 rotor at 4 °C) was used as a source of LCAT and LTP (28). The d > 1.215 g/ml fraction was washed by a second ultracentrifugal run, concentrated 2–3-fold at 4 °C in a Centricon 30 miniconcentrator, dialyzed against phosphate-buffered saline containing 0.01% EDTA, pH 7.4, and stored at −80 °C. The LPDP fractions contained trace amounts of apoAI (2.5–3.4 mg/dl); in some experiments, the contaminating apoAI was removed by immunoaffinity chromatography (29).

Triglyceride-rich particles (TGRP) were isolated from Intralipid 20%, after removal of multilamellar phospholipid (PL) structures. Two milliliters of Intralipid 20% were added to SW 41 polyallomer tubes and overlayered with a KBr/NaCl/Na_2EDTA solution (d = 1.006 g/ml). The tubes were centrifuged for 15 min at 25,000 rpm in a Beckman SW Ti-41 rotor, sliced 1.0 cm from the top and the lipid "cake" resuspended in the d = 1.006 g/ml solution. This procedure was repeated twice. The isolated TGRP were used for incubations within the same day. The original Intralipid had a TG/PL ratio of 43; in the isolated TGRP this ratio was 168, indicating that the bulk of the non-TG-associated PL had been removed.

Reduction and Carboxamidomethylation of HDL_3—HDL_3 (d = 1.125–1.21 g/ml) isolated from three AI_M carriers and from age-sex-matched controls were dialyzed against phosphate-buffered saline and reacted with dithiothreitol (1.2 mg/mg of protein) for 2 h at 37 °C under N_2. A 2-fold excess of iodoacetamide was then added and the mixture allowed to stand for 30 min at 0 °C. After dialysis against a 0.15 M NaCl, 0.01% Na_2EDTA, 0.02% NaN_3 solution, pH 7.4, the mixture was brought to a density of 1.063 g/ml with solid KBr and centrifuged at 43,000 rpm for 18 h in a Beckman Ti-50 rotor. The density of the bottom fraction was raised to 1.21 g/ml and HDL isolated by ultracentrifugation at 40,000 rpm for 48 h. Nomenclature of the different HDL preparations was as follows: (*a*) N-HDL_3, native HDL_3; (*b*) B-HDL_3, "blank" HDL_3 isolated after incubation in the absence of reducing-carboxamidomethylating agents; (*c*) RCM-HDL_3, reduced-carboxamidomethylated HDL_3. The complete reduction of the $apoAI_M$ and -AII dimers in RCM-HDL_3 was checked by sodium dodecyl sulfate-polyacrylamide gel electrophoresis, according to Laemmli (30).

Incubations—All lipoprotein fractions were dialyzed against phosphate-buffered saline just before use. Incubation studies were performed within 4 days from lipoprotein isolation.

In experiments where changes in the plasma HDL subfraction distribution were monitored, aliquots of untreated plasma and of plasma containing 1.5 mM sodium iodoacetate (NaIAc) as an LCAT inhibitor (31) were incubated under N_2 in a shaking water bath at 37 °C for 6 h. After the incubations, very low density lipoprotein and low density lipoprotein were precipitated with $MgCl_2$/phosphotungstic acid from pentuplicate plasma aliquots and lipid concentrations measured in the supernatants (32). HDL subfractions were separated by rate zonal ultracentrifugation (27) and pooled fractions analyzed for lipid and protein content; HDL particle size was determined by GGE. In experiments where changes of HDL size and composition following exposure to TGRP and LPDP were monitored, isolated HDL_3 (2.5 mg/ml protein), TGRP (20.0 mg/ml TG) and LPDP were incubated for 24 h at 37 °C; when specified, LCAT activity was inhibited by adding NaIAc (1.5 mM final concentration). HDL subclasses distribution was analyzed by rate zonal ultracentrifugation and GGE. HDL (d = 1.063–1.21 g/ml) were also isolated by salt density ultracentrifugation and lipid and protein contents measured.

Analyses—Concentrations of total and free cholesterol (TC and FC), TG and PL in plasma and isolated lipoprotein fractions were measured by enzymatic methods (33–35). The CE mass was calculated as (TC − FC) × 1.68. Protein was determined by the method of Lowry *et al.* (36), with bovine serum albumin as a standard. Serum HDL-C levels were measured after precipitation of apoB-containing lipoproteins with dextran/$MgCl_2$ (37). Gel filtration chromatography was carried out on a 1.6 × 95-cm column packed with Bio-Gel A0.5m and equilibrated with 0.15 M NaCl, 0.01% Na_2EDTA, and 0.02% NaN_3, pH 7.4. The α-helical content of isolated HDL_3 was calculated from circular dichroism spectra recorded with a Jasco J500A spectropolarimeter, as described previously (38).

Statistical analyses were carried out by the Student's t test.

RESULTS

Lipoprotein Changes during Incubation of Plasma—Incubations of whole plasma were performed on samples collected from nine different AI_M carriers and from age-sex-matched unaffected subjects from the same kindred (25). The plasma lipid levels in the examined carriers and controls are reported in Table I. There is a 25% decrease in free cholesterol (FC) concentrations in control samples (from 62.2 ± 10.6 to 46.6 ± 9.1 mg/dl) after incubation for 6 h at 37 °C; significantly lower values are found in AI_M subjects: from 53.6 ± 15.3 to 44.8 ± 12.3 mg/dl (−16.5%; $p < 0.01$ *versus* controls). The decrease in free cholesterol during the incubation can be prevented (>97%) in both cases by the LCAT inhibitor NaIAc (controls, 61.7 ± 10.9 mg/dl; AI_M, 51.9 ± 14.8 mg/dl).

Marked changes in the plasma levels of HDL lipids occur during incubation of whole plasma, both in the absence and presence of NaIAc (Table II). The TG and PL contents increase in both conditions, changes being less pronounced in experiments with AI_M plasma; the TG rise is consistently 70% lower than in controls, whereas the PL increase is 47 and 33% lower in AI_M incubations with or without LCAT inhibitor, respectively. HDL-CE increase significantly in the presence of active LCAT, 40% less in AI_M HDL, and decrease, due to a net mass transfer to very low density lipoprotein (15), when LCAT is inhibited; this decrease is 60% lower in

TABLE I

Plasma lipid levels in the studied subjects

Results are the means ± S.D., n = 9. FC, free cholesterol.

	Controls	AI_{Milano}
	mg/dl	
TC	205.2 ± 45.3	152.2 ± 33.3
FC	62.2 ± 10.6	53.6 ± 15.3
TG	123.4 ± 60.9	169.5 ± 83.7
PL	237.6 ± 39.7	206.9 ± 72.1
HDL-C	46.2 ± 10.0	13.5 ± 5.3

TABLE II

Changes in the plasma levels of HDL lipids during incubation of control and AI_{Milano} whole plasma

Plasma samples from nine AI_M carriers and from age-sex-matched unaffected subjects were incubated under N_2 at 37 °C for 6 h both in the presence (+) and absence (−) of 1.5 mM NaIAc as LCAT inhibitor. Before and after the incubations, VLDL + LDL were precipitated with $MgCl_2$/phosphotungstic acid from pentuplicate aliquots and lipid concentrations measured in the supernatants. Differences between incubated and nonincubated samples were calculated and reported as means ± S.D. FC, free cholesterol.

	−NaIAc	+NaIAc
	mg/dl	
Controls		
FC	+0.5 ± 0.2	+2.3 ± 0.8
CE	+15.6 ± 5.3	−11.3 ± 1.7
TG	+10.8 ± 1.5	+8.8 ± 1.0
PL	+32.3 ± 4.7	+16.7 ± 1.7
AI_{Milano}		
FC	+0.3 ± 0.1[a]	+1.8 ± 0.5
CE	+9.7 ± 2.4[b]	−4.3 ± 1.7[c]
TG	+3.2 ± 1.8[c]	+2.2 ± 1.1[c]
PL	+21.7 ± 5.3[c]	+8.9 ± 3.3[c]

[a] $P < 0.05$.
[b] $P < 0.01$.
[c] $P < 0.001$ *versus* controls.

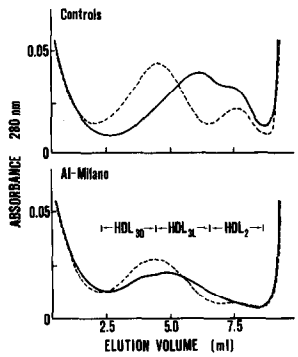


FIG. 1. **Rate zonal ultracentrifugation profiles of HDL subfractions before and after incubation of whole plasma.** Plasma samples from nine AI_M carriers and from age-sex-matched controls were incubated under N_2 at 37 °C for 6 h. Before (--) and after (——) the incubation, HDL subfractions were separated by rate zonal ultracentrifugation (27); the mean tracings from the nine ultracentrifugal separations are reported. Fractions corresponding to HDL_{3D}, HDL_{3L}, and HDL_2 were pooled according to Ref. 2 and analyzed for lipid and protein composition (Table IV).

AI_M plasma. As a result of these changes, the CE/TG weight ratio is reduced from 4.26 to 2.22 in control HDL and from 1.42 to 0.89 in AI_M HDL in the presence of the LCAT inhibitor; it decreases to 3.09 and increases to 1.77, respectively, during incubation with active LCAT.

Changes in HDL subclasses were investigated using two different separation techniques: according to lipoprotein flotation rate by rate zonal ultracentrifugation (2, 27), and according to particle size by GGE (3). Incubation of control plasma, both in the presence and absence of the LCAT inhibitor, results in a pronounced increase in the flotation rate of HDL_3. The HDL_3 peak is shifted toward a lower density, intermediate between HDL_3 and HDL_2 (Fig. 1), the elution volume (V_e) of the HDL_3 peak increasing from 4.3 ± 0.3 to 5.8 ± 0.4 ml (−NaIAc, $p < 0.001$) and to 5.4 ± 0.4 ml (+NaIAc, $p < 0.001$). The flotation rate of HDL_2 does not change, elution volumes being 7.5 ± 0.2, 7.5 ± 0.2, and 7.6 ± 0.2 ml, respectively. The shift of the HDL_3 peak is accompanied by a marked reduction of the 280 nm absorbance in the region of slower-floating HDL_3 particles, also called HDL_{3D} (2), with a concomitant increase in the region of faster-floating HDL_{3L} and of HDL_2 (Fig. 1). Since the AI_M carriers are generally hypertriglyceridemic (39), a plasma sample collected from a fasting patient with type IV hyperlipoproteinemia (TC = 190 mg/dl, TG = 269 mg/dl, HDL-C = 23 mg/dl) was also incubated at 37 °C for 6 h. The changes in the HDL subfraction profile following incubation are similar to those detected in normolipidemic controls: the HDL_2 V_e does not change, the HDL_3 V_e increasing from 4.33 to 6.53 ml (−NaIAc) and to 5.60 ml (+NaIAc).

When plasma samples from AI_M subjects are incubated under identical conditions, only a portion of native HDL_3 (V_e = 3.9 ± 0.2 ml) is converted to faster-floating particles. This results in a bimodal distribution (Fig. 1) of material floating in the HDL_3 density range (−NaIAc: V_e = 4.7 ± 0.2 and 3.7 ± 0.2 ml; +NaIAc: V_e = 5.0 ± 0.2 and 3.6 ± 0.2 ml). These changes in HDL subfraction distribution are independent from the plasma TG and HDL-C levels in the individual carriers.

To quantitate the modifications in the HDL subclasses following incubation, fractions from the rate zonal gradients were pooled according to Ref. 2 in HDL_{3D}, HDL_{3L}, and HDL_2. The mass of these subfractions was calculated, for individual samples, from measurements of their lipid and protein contents. Control plasma incubated in the presence of active LCAT shows a 41% reduction of the HDL_{3D} mass, with a concomitant increase of both HDL_{3L} and HDL_2, by 18 and 25%, respectively (Table III). When LCAT is inhibited, HDL_{3D} are reduced by 47%, HDL_{3L} and HDL_2 increasing by 11 and 26%. All subfractions become TG-enriched during the incubations (Table IV); the PL content decreases in the presence of active LCAT. CE accumulate in all subfractions, significantly only in HDL_{3L}, when LCAT is active and decrease, significantly in HDL_{3D}, in the presence of the LCAT inhibitor.

According to the rate zonal profiles, lower order changes in the levels of HDL subfractions are recorded in incubated AI_M plasma. The HDL_{3D} mass decreases by 15% in the absence of LCAT inhibitor, and by 21% when NaIAc is added to plasma (Table III). These reductions are accompanied by slight, nonsignificant increases in the plasma levels of both HDL_{3L} and

TABLE III

Plasma levels of HDL subfractions (total lipoprotein mass) before and after incubations of control and AI_{Milano} whole plasma

Plasma samples from nine AI_M carriers and from age-sex-matched controls were incubated as reported in Table II. Before and after incubation, HDL were separated by rate zonal ultracentrifugation (Fig. 1) and fractions pooled in slow-floating HDL_3 (HDL_{3D}), fast-floating HDL_3 (HDL_{3L}) and HDL_2, according to Ref. 2. Lipid and protein contents were determined on pooled fractions and lipoprotein mass calculated as the sum of free cholesterol, cholesteryl esters, phospholipids, triglycerides, and protein. Results are the means ± S.D. (n = 9).

	Controls	AI_{Milano}
	mg/dl	
HDL_{3D}		
Nonincubated	93.9 ± 19.5	57.9 ± 8.2
−NaIAc	55.7 ± 13.8	49.0 ± 7.2
+NaIAc	49.8 ± 13.0	45.8 ± 4.3
HDL_{3L}		
Nonincubated	117.1 ± 24.1	56.9 ± 22.6
−NaIAc	138.2 ± 26.3	63.1 ± 16.6
+NaIAc	129.8 ± 22.8	60.1 ± 19.0
HDL_2		
Nonincubated	63.7 ± 18.4	16.0 ± 4.4
−NaIAc	79.8 ± 19.4	17.3 ± 4.4
+NaIAc	80.3 ± 18.9	18.3 ± 6.0

TABLE IV

HDL subfraction composition (percent of total lipoprotein mass) before and after incubation of control and AI_{Milano} whole plasma

HDL_{3D}, HDL_{3L}, and HDL_2, separated as described in Table III, were analyzed for lipid and protein contents, and percentage lipoprotein composition was calculated. Results are the means ± S.D. ($n = 9$). FC, free cholesterol.

	FC		CE		TG		PL		Protein	
	Control	AI_M	Control	AI_M	Control	AI_M	Control	AI_M	Control	AI_M
HDL_{3D}										
Nonincubated	1.7 ± 0.1	1.7 ± 0.3	16.9 ± 0.9	7.6 ± 2.5	2.4 ± 1.3	6.0 ± 0.9	22.0 ± 0.9	25.4 ± 2.5	57.0 ± 1.6	59.3 ± 1.9
−NaIAc	1.6 ± 0.3	1.9 ± 0.3	17.6 ± 0.6	9.7 ± 2.0	4.7 ± 1.5[a]	6.6 ± 2.9	19.4 ± 2.1[a]	24.6 ± 1.1	56.7 ± 3.0	57.2 ± 1.9
+NaIAc	2.0 ± 0.4	2.4 ± 0.3	15.9 ± 1.1[b]	8.0 ± 2.1	4.1 ± 1.2[a]	6.7 ± 1.6	23.3 ± 3.1	28.8 ± 2.0	54.7 ± 2.9	54.2 ± 3.4
HDL_{3L}										
Nonincubated	2.7 ± 0.3	2.8 ± 0.5	18.8 ± 1.5	9.0 ± 2.9	3.2 ± 1.4	7.5 ± 1.6	26.7 ± 1.3	32.2 ± 2.5	48.6 ± 1.8	48.4 ± 1.3
−NaIAc	2.4 ± 0.5	2.6 ± 0.4	21.3 ± 0.8[a]	12.5 ± 3.0[b]	3.7 ± 1.2	8.3 ± 2.3	24.4 ± 1.6[b]	28.4 ± 1.7[b]	48.3 ± 2.1	47.7 ± 1.3
+NaIAc	3.2 ± 0.4[a]	3.3 ± 0.5[b]	18.5 ± 0.7	8.4 ± 2.7	4.3 ± 1.4[a]	8.5 ± 2.7	28.0 ± 1.3	33.6 ± 1.4	46.2 ± 1.7	46.3 ± 2.0
HDL_2										
Nonincubated	4.4 ± 0.5	3.4 ± 1.0	21.3 ± 2.2	10.6 ± 1.7	4.6 ± 2.5	16.7 ± 5.5	30.1 ± 2.4	31.6 ± 4.4	39.5 ± 5.9	37.7 ± 5.3
−NaIAc	3.4 ± 0.7[a]	2.9 ± 0.9	22.7 ± 3.3	12.8 ± 3.8[a]	5.5 ± 1.7[a]	18.4 ± 3.2	27.4 ± 1.7[a]	28.4 ± 4.2	41.0 ± 2.2	38.3 ± 6.1
+NaIAc	4.6 ± 0.7	3.8 ± 0.9	20.7 ± 1.4	9.5 ± 2.1	5.6 ± 2.8[a]	17.7 ± 1.7	32.0 ± 0.9	32.6 ± 3.2	37.2 ± 1.6	36.4 ± 3.8

[a] $P < 0.05$.
[b] $P < 0.01$ *versus* nonincubated.

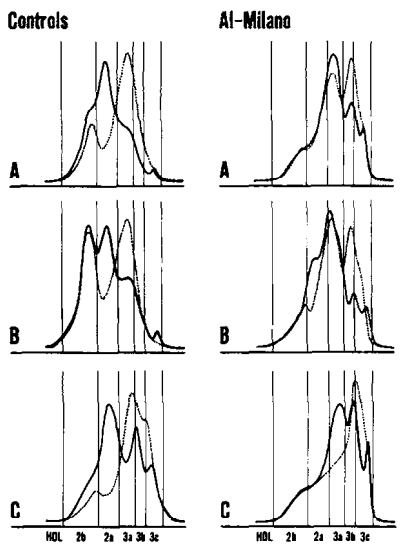


FIG. 2. **GGE profiles of HDL before and after incubation of whole plasma.** Plasma samples from three AI_M carriers (*right, A*, VII-46 (25), TC = 124 mg/dl, TG = 126 mg/dl, HDL-C = 10 mg/dl; *B*, VI-176, TC = 203 mg/dl, TG = 94 mg/dl, HDL-C = 23 mg/dl; *C*, VII-84, TC = 218 mg/dl, TG = 370 mg/dl, HDL-C = 10 mg/dl) and from age-sex-matched controls (*left, A*, TC = 156 mg/dl, TG = 57 mg/dl, HDL-C = 50 mg/dl; *B*, TC = 205 mg/dl, TG = 53 mg/dl, HDL-C = 67 mg/dl; *C*, TC = 190 mg/dl, TG = 269 mg/dl, HDL-C = 23 mg/dl) were incubated under N_2 at 37 °C for 6 h. The $d < 1.20$ g/ml fractions were isolated by ultracentrifugation before (——) and after (——) the incubations and the HDL particle size distribution analyzed by nondenaturing polyacrylamide gradient gel electrophoresis (3). The same amount of HDL protein was applied to the gel.

HDL_2. All native HDL subfractions from the AI_M carriers are TG-enriched compared to controls; a further, nonsignificant rise of the TG content is noted after the incubation (Table IV). CE contents are about one-half of those of control subfractions; they increase, particularly in faster-floating fractions, during incubation with active LCAT, and do not change in the presence of NaIAc.

Fig. 2 (*left*) shows the GGE profiles of HDL isolated by ultracentrifugation of control plasma, following incubation at 37 °C or storage at 4 °C for 6 h. Incubation dramatically alters the particle size distribution of HDL (5, 6): in all subjects there is a marked reduction in the HDL_{3a} peak together with the appearance of large HDL_{2a} species (9.54 ± 0.06 nm); the particle sizes of HDL_{2b} and of the remaining HDL_{3a} do not change (from 10.35 ± 0.09 to 10.36 ± 0.11 nm and from 8.73 ± 0.12 to 8.67 ± 0.08 nm, respectively). A small, distinct peak, migrating in the HDL_{3c} size interval (7.63 ± 0.04 nm), become evident. Incubation of a plasma sample from a hypertriglyceridemic patient with a low HDL-C level results in similar modifications of the HDL profile (Fig. 2*C*, *left*).

The HDL pattern in the AI_M carriers differ from controls for the presence of a distinct HDL subpopulation migrating within the HDL_{3b} interval, in addition to the normal HDL_{3a}; the HDL_{2b} peak is markedly reduced compared to controls (23) (Fig. 2, *right*). In all the AI_M carriers, incubation results in a partial reduction of material migrating within the HDL_{3b} size interval; this is associated with a slight increase in the HDL_{3a} peak and with the appearance of small particles, in the HDL_{3c} size interval (7.62 ± 0.05 nm). A distinct HDL_{2a} peak never appears, although some HDL_{2a} particles are formed, especially in normotriglyceridemic female carriers with relatively high HDL-C levels (Fig. 2*B*, *right*). The particle sizes of both HDL_{3a} (8.84 ± 0.15 nm) and HDL_{3b} (8.15 ± 0.05 nm) do not change (8.82 ± 0.17 and 8.09 ± 0.09 nm, respectively).

Changes in the HDL_3 size distribution were also analyzed by gel filtration (Fig. 3). HDL_3 were isolated by preparative ultracentrifugation from a control and from an AI_M sample, before and after incubation, and filtered through a calibrated agarose column. Control, nonincubated HDL_3 elute as a symmetrical peak, with a V_e of 131.7 ml, corresponding to a particle diameter of 9.4 nm; after incubation, the HDL_3 peak widens, with the appearance of particles eluting at lower volumes, and the peak V_e is shifted to 129.5 ml (particle diameter = 9.9 nm). Nonincubated AI_M HDL_3 are smaller and more heterogeneous compared to control HDL_3; the peak is wider and elutes at 135.9 ml (8.6 nm). The shape and the V_e of the HDL_3 peak are almost unchanged after incubation (Fig. 3). These findings are consistent with the GGE data.

Incubation of HDL_3 with TGRP—The anomalous behavior of HDL observed during the incubation of AI_M whole plasma may be due either to the low HDL levels, or to a deficiency of factors involved in HDL particle conversion, or to structural alterations of HDL and/or "acceptor" lipoproteins. A partial LCAT deficiency state in AI_M carriers, due to low plasma levels of a functional enzyme, has been recently reported (24);

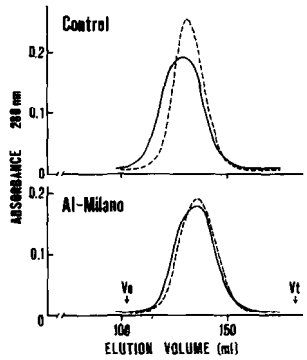


FIG. 3. **Gel filtration chromatography of HDL before and after incubation of whole plasma.** Plasma samples from a control subject and from an AI_M carrier were incubated under N_2 at 37 °C for 6 h. HDL (d = 1.063–1.21 g/ml) were isolated by ultracentrifugation before (--) and after (——) the incubation and filtered through a calibrated Bio-Gel A0.5m column (1.6 × 95 cm) equilibrated with 0.15 M NaCl, 0.01% Na_2EDTA, 0.02% NaN_3, pH 7.4.

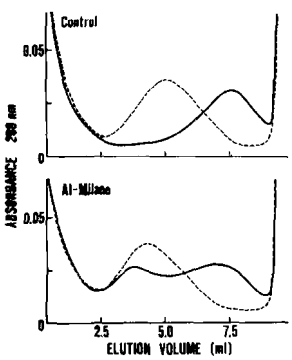


FIG. 4. **Rate zonal ultracentrifugation profiles of HDL before and after incubation of purified HDL_3 with TGRP and LPDP.** HDL_3 (d = 1.125–1.21 g/ml) purified from five AI_M carriers and from age-sex-matched controls were incubated with an LPDP and with TGRP, isolated from Intralipid, at 37 °C for 24 h. Before (---) and after (——) the incubation, HDL were separated by rate zonal ultracentrifugation and the mean tracings reported.

furthermore, the plasma content of CETP is also reduced in these subjects.[2] *In vitro* experiments were thus designed to evaluate the role of the abnormal lipid/apolipoprotein composition of AI_M HDL in determining the impaired HDL particle conversion detected in the AI_M plasma. Incubations were carried out by mixing an LPDP, prepared from a control plasma, with purified HDL_3 (from control and AI_M subjects) and TGRP, to yield a HDL_3-protein/TGRP-TG ratio of 1/8. In some experiments LCAT activity was inhibited by adding 1.5 mM NaIAc. Nonincubated (24 h at 4 °C) and incubated (24 h at 37 °C) mixtures were analyzed by rate zonal ultracentrifugation and GGE. When the trace amounts of free apoAI present in the LPDP fraction were removed by immunoaffinity chromatography, the HDL conversion pathway, both with control and AI_M HDL_3, was not affected (data not shown).

Following the incubation, control HDL_3 exhibits a pronounced increase in the flotation rate; the HDL_3 peak is shifted to a position close to HDL_2 (Fig. 4). In five experiments, using different HDL_3 preparations, the elution volume of the rate zonal peak increases from 4.72 ± 0.27 to 7.25 ± 0.48 ml (+NaIAc) and to 7.78 ± 0.35 ml (−NaIAc) (the V_e of HDL_2 from a control population of 50 males was of 8.01 ± 0.28 ml). Incubation of purified AI_M HDL_3 under identical conditions results in a broadening of the HDL_3 peak, with the appearance of two distinct subclasses of particles, one (+NaIAc: V_e = 4.37 ± 0.35 ml; −NaIAc: V_e = 3.84 ± 0.19 ml) with a flotation rate lower than nonincubated HDL_3 (V_e = 4.46 ± 0.29 ml) and one floating faster (+NaIAc: V_e = 6.94 ± 0.63 ml; −NaIAc: V_e = 7.26 ± 0.47 ml) than native HDL_3 (Fig. 4). The prolongation of the incubation up to 48 h does not result in further changes in the rate zonal profile (not shown).

Changes in HDL particle size following incubation with TGRP and LPDP are reported in Fig. 5. Control HDL_3 consists of a single population of particles, migrating within the HDL_{3a} interval (mean diameter = 8.64 ± 0.19 nm). After incubation, the HDL_{3a} material almost disappears and HDL_{2a} particles (9.51 ± 0.04 nm) largely predominate (Fig. 5*A*, *left*). A small, distinct peak, migrating in the HDL_{3c} size interval (7.29 ± 0.08 nm), is sometime detectable. HDL_3 from the AI_M carriers are more heterogeneous, compared to control particles; two different subpopulations can be separated by GGE: HDL_{3a} (8.73 ± 0.11 nm) and HDL_{3b} (8.06 ± 0.02 nm). In three subjects the HDL_{3b} peak is predominant (pattern I in Ref. 23); in the other two, the HDL_{3a} and HDL_{3b} particles equally contribute to total HDL (pattern II in Ref. 23). Following incubation, a reduction of material in the HDL_{3a} region is noted, with the appearance of larger species (9.31 ± 0.28 nm), similar to those found after incubation of control HDL_3. Differently from controls, a distinct HDL_{3b} peak (7.93 ± 0.08 nm) is still detectable after incubation (Fig. 5*A*, *right*). LCAT inhibition by NaIAc minimally affects the changes in HDL particle size distribution, both with control and AI_M HDL.

The total HDL fraction (d = 1.063–1.21 g/ml) from the five nonincubated and incubated mixtures was also isolated by salt density ultracentrifugation, and the chemical composition determined (Table V). Incubation of control HDL_3 results in a marked reduction of their CE with a concomitant increase of TG contents; similar, but less pronounced changes are observed after incubation of AI_M HDL_3. The free cholesterol and protein contents decrease slightly, whereas PL are only marginally affected, both in control and AI_M HDL_3. As a result of these changes, the CE/TG ratio decreases from 4.61 ± 2.50 to 0.51 ± 0.14 in control HDL_3 and from 1.47 ± 0.35 to 0.23 ± 0.07 in AI_M HDL_3.

Incubation of Reduced-Carboxamidomethylated HDL_3 with TGRP—To verify if the $apoAI_M$ homo- and heterodimers are involved in the impaired HDL particle conversion observed

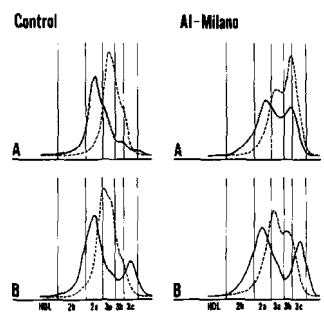


FIG. 5. **GGE profiles of HDL before and after incubation of native (*A*) and reduced-carboxamidomethylated (*B*) HDL_3 with TGRP and LPDP.** HDL_3 (d = 1.125–1.21 g/ml), purified from three AI_M carriers and from age-sex-matched controls, were reduced with 40 mM dithiothreitol and carboxamidomethylated with a 2-fold excess of iodoacetamide. Native (*A*) and reduced-carboxamidomethylated HDL_3 (*B*) were incubated with a LPDP and with TGRP, isolated from Intralipid, at 37 °C for 24 h. Before (---) and after (——) the incubation, the HDL particle size distribution was analyzed by nondenaturing polyacrylamide gradient gel electrophoresis. Typical profiles from one AI_M carrier (*right*) and the respective control (*left*) are shown. The same amount of HDL_3 protein was applied to the gel.

[2] Y. L. Marcel, and G. Franceschini, unpublished observations.

TABLE V

Chemical composition of HDL isolated before and after incubation of HDL_3 with LPDP and TGRP

HDL_3 (d = 1.125–1.21 g/ml) were purified from five AI_M carriers and from age-sex-matched controls, and incubated with an LPDP and with TGRP purified from Intralipid, to yield a HDL_3-protein/TGRP-TG ratio of 1:8. Incubations were carried out under N_2 at 37 °C for 24 h. Before and after the incubation, total HDL (d = 1.063–1.21 g/ml) were purified by salt density ultracentrifugation, and lipid-protein composition determined. Results are the means ± S.D., n = 5. FC, free cholesterol.

	Control		AI_{Milano}	
	Nonincubated	Incubated	Nonincubated	Incubated
	% of weight			
FC	2.3 ± 0.6	1.1 ± 0.3	2.6 ± 0.2	0.8 ± 0.3
CE	17.6 ± 1.3	8.9 ± 1.2	9.9 ± 1.1	4.0 ± 1.0
TG	4.6 ± 1.3	17.9 ± 3.3	6.9 ± 1.1	17.3 ± 1.1
PL	25.3 ± 0.6	25.1 ± 0.6	32.4 ± 2.4	31.0 ± 1.2
Proteins	50.2 ± 0.3	47.2 ± 1.5	48.3 ± 1.7	47.0 ± 1.8

TABLE VI

Chemical composition of HDL isolated before and after incubation of native and RCM-HDL_3 with LPDP and TGRP

HDL_3 were purified from three AI_M carriers and from age-sex-matched controls, reduced with 40 mM dithiothreitol and carboxamidomethylated with a 2-fold excess of iodoacetamide. Reisolated HDL (d = 1.063–1.21 g/ml) were incubated with LPDP and TGRP at 37 °C for 24 h. Before and after the incubation, total HDL (d = 1.063–1.21 g/ml) were purified by salt density ultracentrifugation, and lipid-protein composition was determined. Results are the means ± S.D., n = 3. FC, free cholesterol.

	Native		RCM	
	Nonincubated	Incubated	Nonincubated	Incubated
	% of weight			
Controls				
FC	2.2 ± 0.1	1.0 ± 0.1	2.2 ± 0.1	1.0 ± 0.1
CE	17.0 ± 1.2	10.4 ± 0.1	17.8 ± 1.1	8.2 ± 0.4
TG	3.0 ± 1.0	13.5 ± 0.1	3.9 ± 1.3	16.6 ± 2.4
PL	26.6 ± 1.0	27.4 ± 0.4	27.6 ± 1.1	28.2 ± 1.8
Protein	49.8 ± 1.1	47.7 ± 0.3	48.0 ± 1.0	46.0 ± 0.9
AI_{Milano}				
FC	2.4 ± 0.2	0.7 ± 0.1	2.6 ± 0.5	0.8 ± 0.1
CE	9.5 ± 1.4	5.7 ± 2.3	9.1 ± 1.8	4.5 ± 1.0
TG	7.3 ± 2.0	15.8 ± 2.6	7.5 ± 1.0	17.6 ± 0.5
PL	31.6 ± 0.5	32.1 ± 1.1	32.3 ± 0.2	31.5 ± 1.0
Protein	48.5 ± 1.9	45.9 ± 1.3	48.0 ± 2.0	45.6 ± 0.4

in AI_M plasma and in incubations with purified AI_M HDL_3, the AI_M dimers were reduced and carboxamidomethylated. A similar treatment of control HDL_3 resulted in the reduction-carboxamidomethylation of the apoAII dimer. The particle size distribution and the lipid-protein composition of blank HDL_3 and of native HDL_3 were similar, indicating that the incubation necessary for the chemical modification of disulfide bonds does not affect lipoprotein structure and composition. By contrast, the size distribution of HDL_3 was slightly affected by reduction-carboxamidomethylation (Fig. 5*B*), the size of both control and AI_M HDL_{3a} increasing from 8.59 ± 0.05 to 8.92 ± 0.08 nm (control) and from 8.68 ± 0.04 to 9.03 ± 0.19 nm (AI_M). However, the composition of RCM-HDL_3 did not differ from that of native lipoproteins (Table VI). The secondary structure of the apolipoprotein molecules in the HDL particles was also unaffected by the reduction-carboxamidomethylation procedure, the percentage of α-helix structure varying between 72.8 ± 3.6% (N-HDL_3) and 74.7 ± 2.8% (RCM-HDL_3) in control HDL_3 and between 74.3 ± 3.4% (N-HDL_3) and 73.7 ± 0.1% (RCM-HDL_3) in AI_M HDL_3.

Reduction and carboxamidomethylation of both control and AI_M HDL_3 dramatically affects their behavior following incubation with TGRP and LPDP (Fig. 5*B*). This is entirely due to the chemical modification of the apolipoprotein molecules, since the conversion pattern of B-HDL_3 does not differ from that of N-HDL_3 (data not shown). After incubation of control RCM-HDL_3, typical HDL_{2a} particles (9.54 ± 0.08 nm) become predominant, as with native HDL_3; an additional, distinct and pronounced peak, migrating in the HDL_{3c} (7.64 ± 0.06 nm) size interval can be noted (Fig. 5*B*, *left*). Different from what was observed with native HDL_3, RCM-HDL_3 from AI_M subjects behave like control RCM-HDL_3: following incubation, the HDL_{2a} peak (9.47 ± 0.08 nm) becomes predominant, with an additional distinct peak appearing in the HDL_{3c} size interval (7.66 ± 0.03 nm) (Fig. 5*B*, *right*). It thus seems that reduction and carboxamidomethylation of the apoAI_M dimers in the HDL_3 transforms these "functionally abnormal" particles into "functionally normal" lipoproteins.

Reduction and carboxamidomethylation slightly affects the compositional modifications of HDL induced by incubation with TGRP and LPDP (Table VI); following incubation, the CE/TG ratio decreases from 4.79 ± 1.72 to 0.49 ± 0.05 in control RCM-HDL_3 and from 1.21 ± 0.15 to 0.24 ± 0.07 in AI_M RCM-HDL_3.

DISCUSSION

Plasticity of the HDL system is a key factor in the modulation of the processes of tissue lipid removal (40). These involve uptake of cellular cholesterol by small, apoAI-containing HDL (10, 11), esterification by LCAT (9), and transfer of the newly formed CE from HDL to lower density lipoproteins (15), responsible for the disposal of CE by way of high affinity receptors in the liver (41). Metabolic disorders affecting these mechanisms are associated with alterations in the structure of HDL (17–19) and, eventually, with an elevated risk of arterial disease (40, 42). Carriers of the apoAI_M mutant show multiple abnormalities in HDL (20–24), not leading, however, to clinically significant vascular alterations (25). Lipid binding experiments showed an increased flexibility of AI_M, compared to normal AI, in the interaction with lipids (38), whereas structural analyses revealed the presence of abnormal HDL particles in carriers of the mutant (22, 23). In the present study we evaluated the *in vitro* metabolic behavior of these abnormal HDL.

In vitro experiments provide a versatile and reliable model for investigating the dynamics of HDL modifications (5–8). As described in previously reported studies (5, 16, 43), dramatic changes in HDL lipids occur during plasma incubations, both in the presence and absence of NaIAc, used as an LCAT inhibitor. CE formation, CE for TG exchange between HDL and lower density lipoproteins and HDL uptake of PL and free cholesterol were, however, generally reduced in AI_M plasma. This is, at least in part, attributable to the partial LCAT deficiency (24) with low plasma CETP levels[2] in the AI_M carriers, and results in a substantially reduced plasticity of AI_M HDL. Thus, in the case of normal HDL, there was a dramatic fall in the density of HDL_3, floating now in the HDL_{3L} position, while only a fraction of AI_M HDL_3 was converted to faster-floating particles. These findings were confirmed by size distribution studies. Incubation led to a predominance of HDL_{2a} particles in control plasma, indicating that *in vitro* HDL is normally capable of enlarging its size (44). In incubated AI_M plasma only a partial redistribution of material in the HDL_3 size range occurred, with minimal formation of HDL_2.

The abnormal behavior of the HDL system in incubated AI_M plasma may be secondary to (*a*) the reported deficiencies

of factors involved in HDL conversion, (*b*) an imbalance between CE donor and CE acceptor lipoproteins, (*c*) structural alterations in these same lipoproteins. An artificial model was therefore designed, in which HDL_3 were isolated from both control and AI_M subjects and incubated with a lipoprotein depleted fraction, together with TG-rich particles purified from Intralipid, and freed of excess PL. In this way it was possible to evaluate whether HDL itself might influence lipoprotein particle conversion. This system proved extremely informative since, after incubation, HDL_3 particles from control subjects increased in size, moving to the HDL_{2a} interval, *i.e.* similar to what was observed in incubated whole plasma. In the case of the AI_M HDL, the majority of material in the HDL_{3b} region was not modified after incubation with TG-rich particles. On the other hand, particles in the HDL_{3a} range nearly disappeared, with the formation of larger HDL_{2a}, in a manner similar to that seen in control HDL. There is thus evidence in AI_M HDL of a fraction that follows a conversion pathway similar to normal HDL, whereas the small, abnormal HDL_{3b} are more resistant to these transformations; this is possibly a consequence of the fact that the AI_M carriers are heterozygotes for the condition (25), having normal HDL, containing normal apoAI, and abnormal HDL, containing the various molecular forms of the mutant apolipoprotein (22).

The abnormal behavior of the AI_M HDL_3 was detected in carriers with widely disperse lipid/lipoprotein levels (Fig. 2), and when incubating HDL fractions of different lipid composition; it appears therefore to be independent of the altered lipid contents of AI_M HDL_3 (12, 45). To verify if the presence of $apoAI_M$ homo- and heterodimers (21) is thus responsible for the impaired conversion of AI_M HDL, these were reduced and carboxamidomethylated, leading to the conversion of the disulfide-linked forms of $apoAI_M$ into their normal counterparts, *e.g.* monomeric AI and AII. This procedure minimally affected lipoprotein structure and composition, but markedly altered the behavior of both control and AI_M HDL_3 during incubation with TG-rich particles. Following incubation with control RCM-HDL_3, both large HDL_{2a} and small HDL_{3c} particles were formed. The mechanism(s) responsible for the formation of the small HDL, not observed during incubation of native HDL_3, are at present under investigation; it can be noted that a similar HDL behavior has been shown following incubation of control plasma in the presence of a reducing agent, 2-mercaptoethanol (6). Even more important, incubation of AI_M RCM-HDL_3 resulted in the formation of both large HDL_{2a} and small HDL_{3c}, as with control RCM-HDL_3; the impaired behavior of native AI_M HDL_3 was thus normalized after reduction-carboxamidomethylation, *i.e.* after conversion of the dimeric forms of $apoAI_M$ into their normal counterparts. Interestingly, the chemical modification had minimal effects on the lipid movements between the various lipoproteins. All these data suggest that, in addition to cholesterol esterification and lipid exchanges (9, 15), and to the activities of lipases (14, 16) and conversion proteins (8), apolipoprotein structure and composition markedly influences the interconversion of HDL. This is in agreement with recent experiments on model lipoproteins, showing that the apolipoprotein content is a key factor in the formation of mature HDL particles (44, 46). In the AI_M carriers, the disulfide-linked forms of the mutant apolipoprotein seem to confer an increased metabolic stability to HDL particles. This is consistent with the recently described delayed *in vivo* catabolism of the AI_M dimers in the carriers (47). These and previous (20–24) studies indicate that a complex derangement of the HDL system, involving apolipoprotein structure, lipoprotein functionality, enzyme activities, and possibly affecting the interaction between HDL and TG-rich lipoproteins (23) is present in the AI_M carriers, making the AI_M condition a unique model for studying the various processes determining the functionality of the entire HDL system.

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